LAB 05 SOLUTION

Build, Test, and Package Applications with IntelliJ IDEA and Docker

Estimated duration: 3 hours

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Course / Section	_____DevOps___ Section A_____
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Devops lab directory URL

https://github.com/sumairamalik6868/DevOps/tree/main/Labs/Lab05

Lab Overview

The purpose of this lab is to use IntelliJ IDEA in the supplied Docker-based desktop environment to build, test, run, and package Java and Node.js applications. The lab covers Maven, Gradle, npm, webpack, and Docker image creation.

Learning Objectives

- Launch IntelliJ IDEA through the supplied Docker environment and clone existing projects.
- Add and synchronize Elasticsearch 7.0.0 dependencies in Maven and Gradle projects.
- Build, test, package, and run Java applications, and identify their generated JAR files.
- Install Node.js and npm, install dependencies, run tests, and create an npm package.
- Resolve the Node.js port conflict by using port 4000.
- Build and inspect the Node.js API bundle generated by webpack.
- Build and verify a Docker image tagged lab5:1.0.

Environment and Command Locations

Use the Ubuntu host terminal to clone the IntelliJ Docker setup repository and run setup.sh. Use the Terminal tool window inside IntelliJ for Maven, Gradle, Node.js, npm, webpack, and Docker commands as specified by the lab. The required terminal identity is sumairamalik6868@ubuntu.

All project clones are stored under /config/IdeaProjects. The submission is stored in the DevOps GitHub repository under Labs/Lab05, with screenshots in Labs/Lab05/screenshots.

Task 1 — Clone and Launch IntelliJ IDEA

The supplied IntelliJ Docker repository provides a browser-accessible desktop containing IntelliJ IDEA. The environment is started with Docker Compose on the Ubuntu host.

Work performed / key points

- Clone https://github.com/WaqasSaleem97/Intellij-Idea-docker.git on the Ubuntu host and enter the repository directory.
- Run bash setup.sh and verify the intellij service with docker compose ps. The desktop uses port 3001.
- Open https://localhost:3001 in the host browser and sign in with the setup browser account.
- Launch IntelliJ IDEA and complete its first-run setup. Keep the required Ubuntu identity visible in the evidence.

Evidence — insert your screenshot below each item

intellij_repo_clone.png — Clone command and successful repository download.

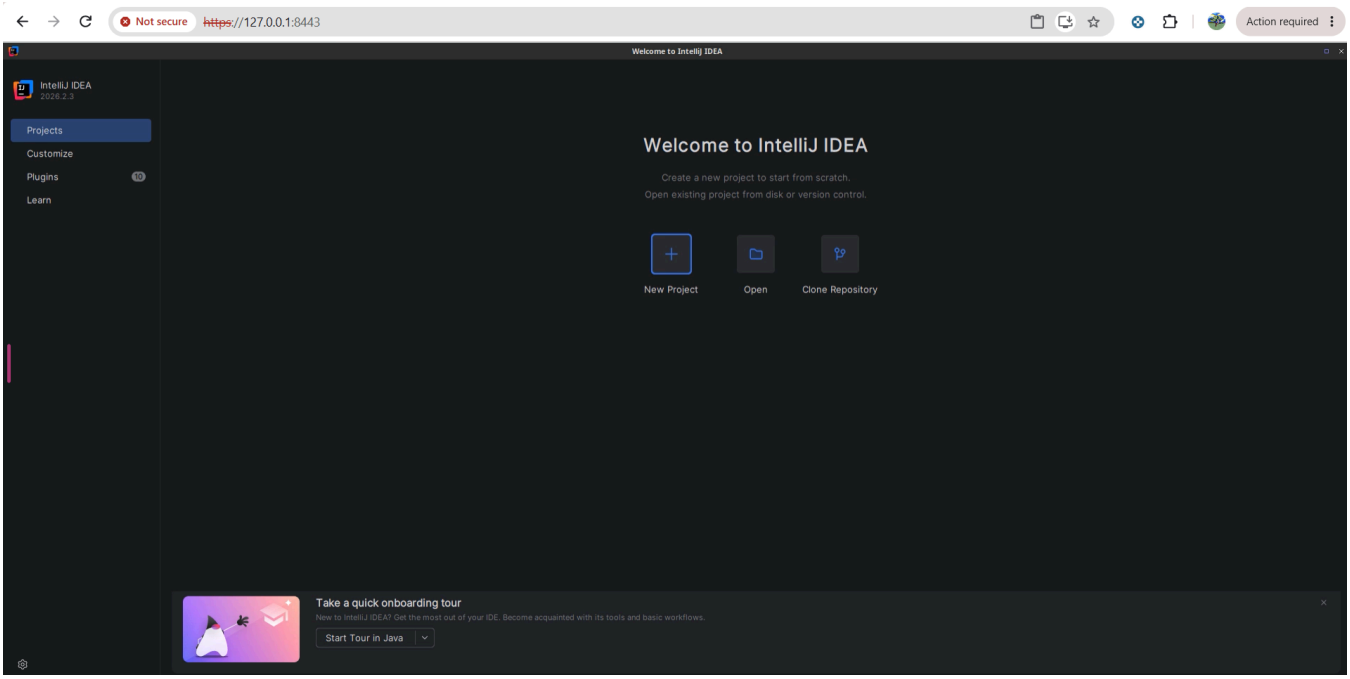
```
sumairamalik6868@ubuntu:~/Intellij-Idea-docker$ cd ~/Intellij-Idea-docker
git remote -v
ls
origin  https://github.com/WaqasSaleem97/Intellij-Idea-docker.git (fetch)
origin  https://github.com/WaqasSaleem97/Intellij-Idea-docker.git (push)
Dockerfile  browser.sh    config            init-terminal.sh       keys       reset-login.sh  root      tests
README.md   compose.yaml  idea-launcher.sh  intellij-idea.desktop  remove.sh  resources.sh    setup.sh
sumairamalik6868@ubuntu:~/Intellij-Idea-docker$
```

intellij_running.png — Running intellij service and desktop port 3001; do not show the password.

```
[+] Building 1/1
 ✔intellij-idea-docker-intellij  Built                                                                    0.0s
[+] Running 2/2
 ✔Network intellij-idea-docker_default       Created                                                      0.3s
 ✔Container intellij-idea-docker-intellij-1  Started                                                      4.2s
NAME                          IMAGE                         COMMAND   SERVICE    CREATED         STATUS         PORTS
intellij-idea-docker-intellij-1   intellij-idea-docker-intellij   "/init"   intellij   4 seconds ago   Up 2 seconds   0.0.0.0:3000-3001->3000-30
01/tcp, [::]:3000-3001->3000-3001/tcp

Browser username configured in the container: admin
Open https://localhost:3001 in your host computer browser.
Use the browser password from setup; the Ubuntu SSH/sudo password is separate.
sumairamalik6868@ubuntu:~/Intellij-Idea-docker$
```

intellij_desktop.png — IntelliJ desktop / Welcome screen with the Ubuntu identity visible.



Task 2 — Build and Run the Maven Project

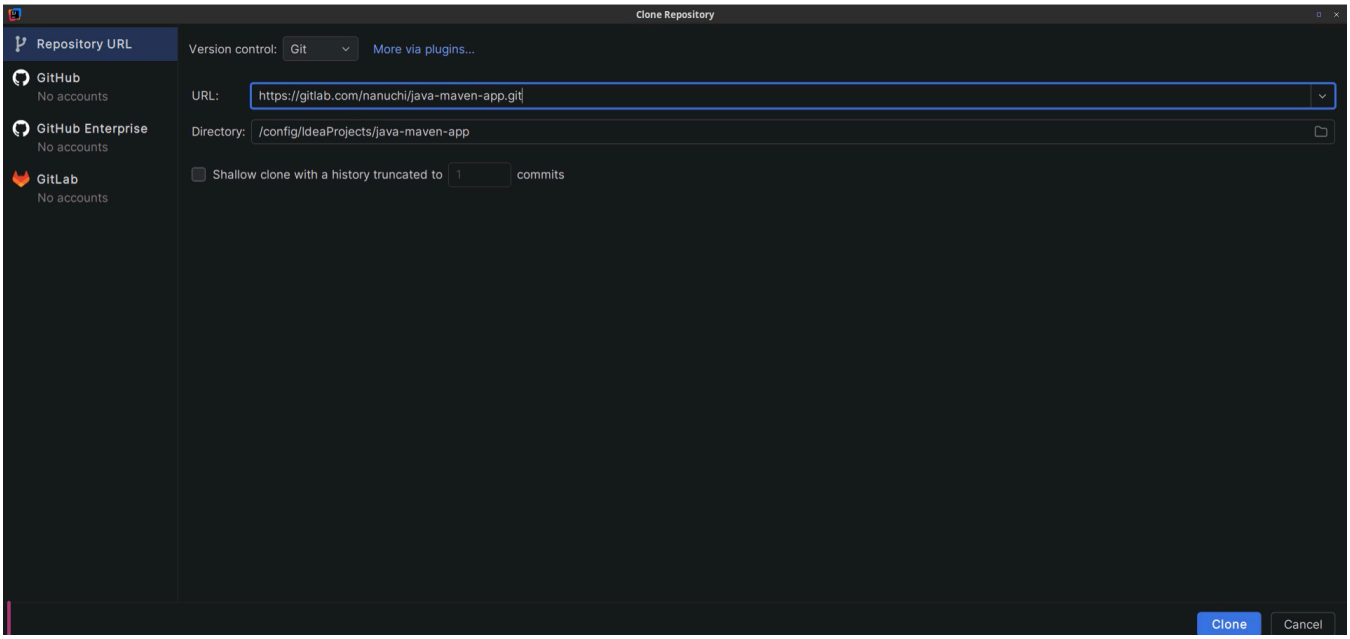
The Maven sample is cloned from https://gitlab.com/nanuchi/java-maven-app.git into /config/IdeaProjects/java-maven-app. Maven uses pom.xml to describe the project and its dependencies.

Work performed / key points

- Install Maven in IntelliJ's terminal and verify it with mvn --version.
- Run mvn package and confirm BUILD SUCCESS.
- Add org.elasticsearch:elasticsearch:7.0.0 with compile scope inside the existing dependencies section of pom.xml. Synchronize Maven changes.
- Run mvn package again, then run mvn test. The supplied sample may report No tests to run with BUILD SUCCESS; record the actual output.
- Locate target/java-maven-app-1.1.0-SNAPSHOT.jar in the Project pane.
- Run java -jar target/java-maven-app-1.1.0-SNAPSHOT.jar. After startup, stop it with Ctrl+C and close the project.

Evidence — insert your screenshot below each item

maven_project_clone.png — Clone dialog showing the Maven repository URL and destination.



maven_installed.png — Maven installation and version output.

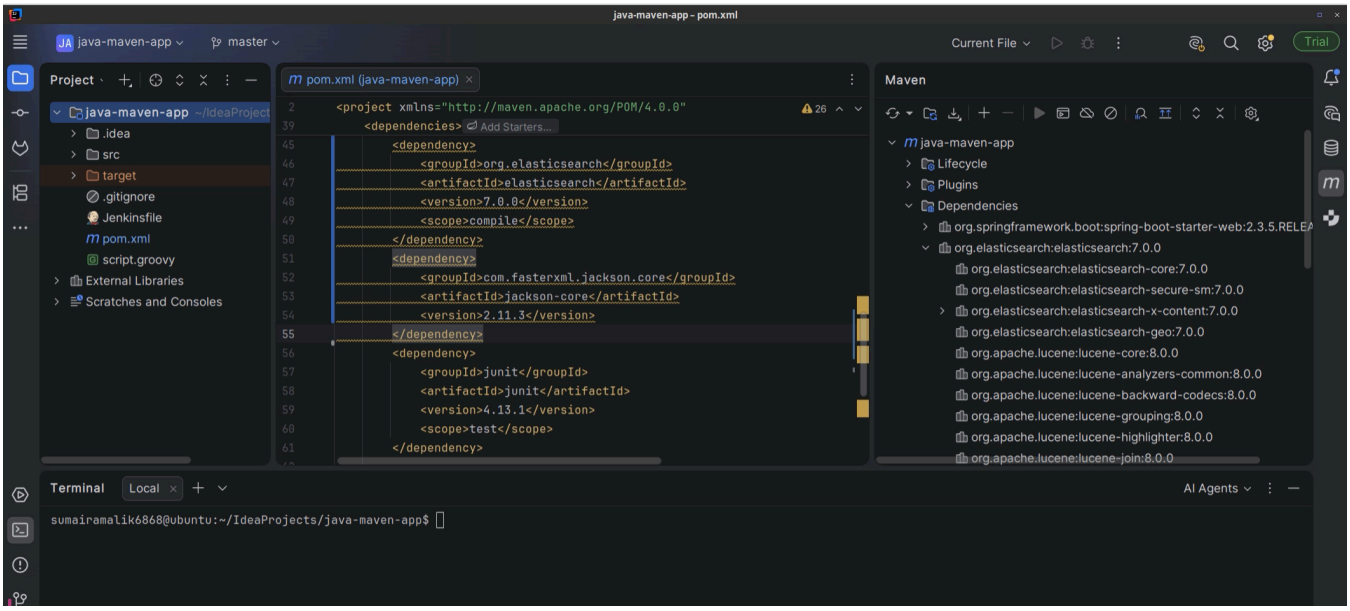


maven_package.png — Initial mvn package command and BUILD SUCCESS.

```
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-classworlds/2.6.0/plexus-classworlds-2.6.0.jar (53 kB at 13 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.2.1/plexus-utils-3.2.1.jar (262 kB at 62 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/sisu/org.eclipse.sisu.inject/0.3.4/org.eclipse.sisu.inject-0.3.4.jar (379 kB at 81 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/net/java/dev/jna/jna/5.5.0/jna-5.5.0.jar (1.5 MB at 96 kB/s)
[INFO] Replacing main artifact with repackaged archive
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  01:03 min
[INFO] Finished at: 2026-11-03T13:36:45+05:00
[INFO] ------------------------------------------------------------------------

sumairamalik6868@ubuntu:~/IdeaProjects/java-maven-app$
```

maven_elasticsearch_dependency.png — pom.xml dependency and synchronized Elasticsearch 7.0.0 entry.



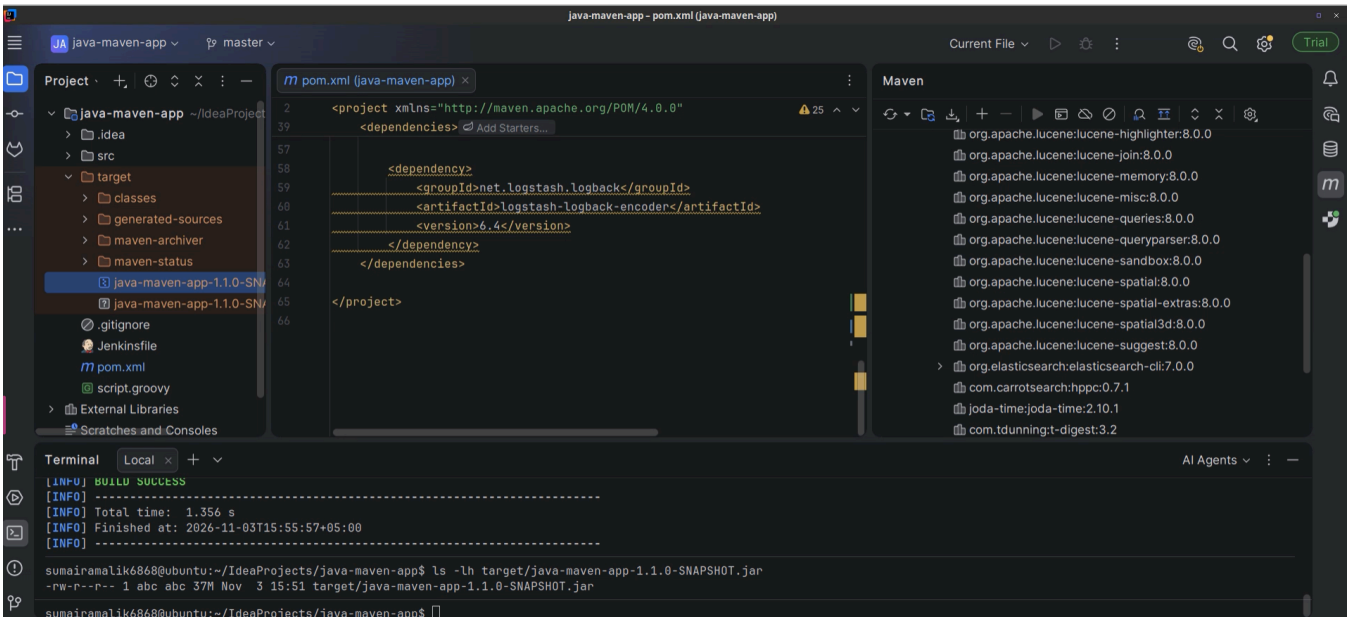
maven_repackage.png — Second successful mvn package after adding the dependency.

```
[INFO] Deleting /config/IdeaProjects/java-maven-app/target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.6.0:compile (default-compile) @ java-maven-app ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /config/IdeaProjects/java-maven-app/target/classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /config/IdeaProjects/java-maven-app/src/test/resources
[INFO]
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.1.2:jar (default-jar) @ java-maven-app ---
[INFO] Building jar: /config/IdeaProjects/java-maven-app/target/java-maven-app-1.1.0-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:2.3.5.RELEASE:repackage (default) @ java-maven-app ---
[INFO] Replacing main artifact with repackaged archive
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.991 s
[INFO] Finished at: 2026-11-03T15:51:16+05:00
```

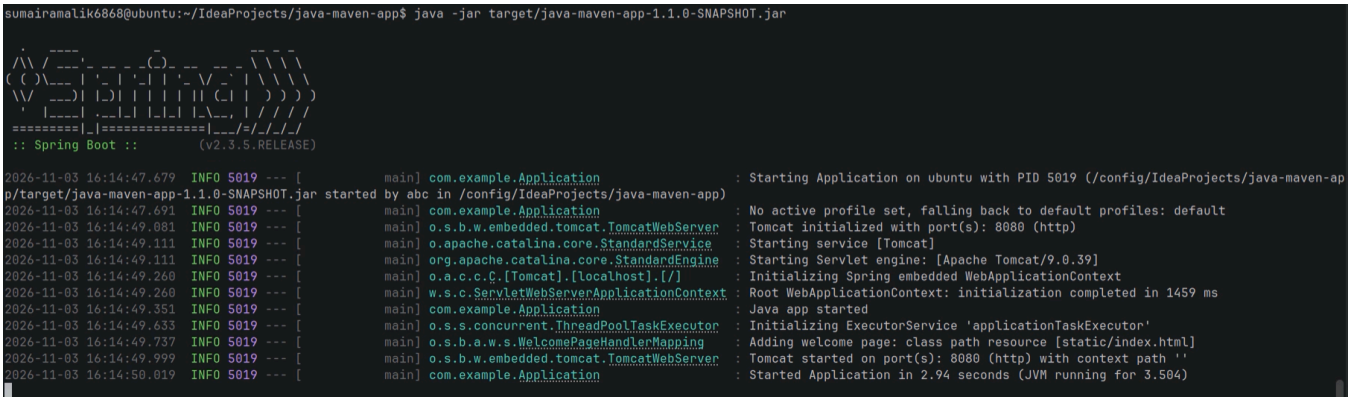
maven_test.png — mvn test output and build result.

```
sumairamalik6868@ubuntu:~/IdeaProjects/java-maven-app$ mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:java-maven-app >--------------------
[INFO] Building java-maven-app 1.1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.6.0:compile (default-compile) @ java-maven-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /config/IdeaProjects/java-maven-app/src/test/resources
[INFO]
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.321 s
[INFO] Finished at: 2026-11-03T15:35:39+05:00
[INFO] ------------------------------------------------------------------------
```

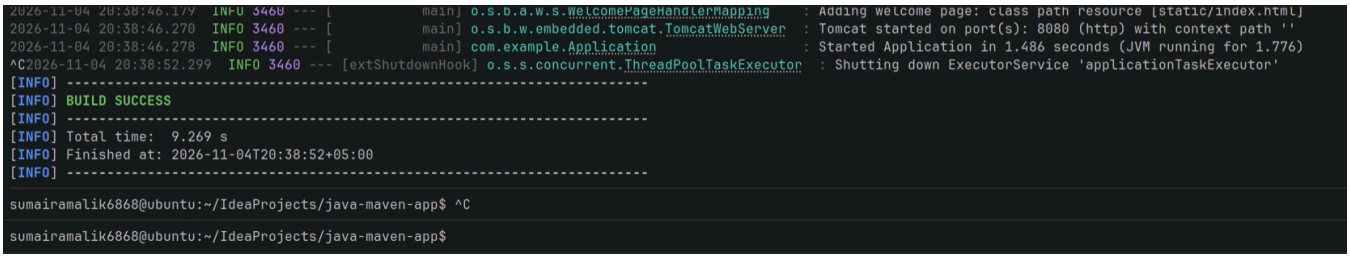
maven_jar.png — target directory and exact JAR filename.



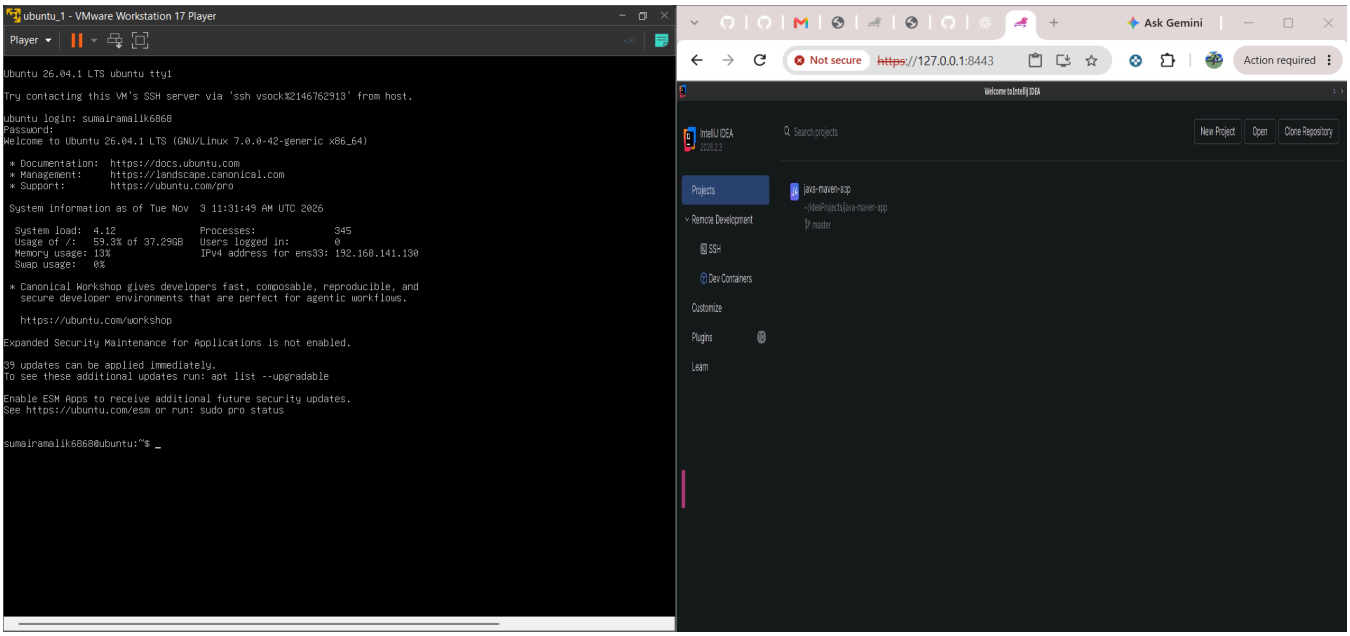
maven_running.png — JAR command and successful application startup.



maven_stopped.png — Ctrl+C / shutdown output and returned terminal prompt.



maven_project_closed.png — IntelliJ Welcome screen after closing the Maven project.



Task 3 — Build and Run the Gradle Project

The Gradle sample is cloned from https://gitlab.com/nanuchi/java-app.git into /config/IdeaProjects/java-app. The project includes the Gradle Wrapper, which runs the project-specified Gradle version.

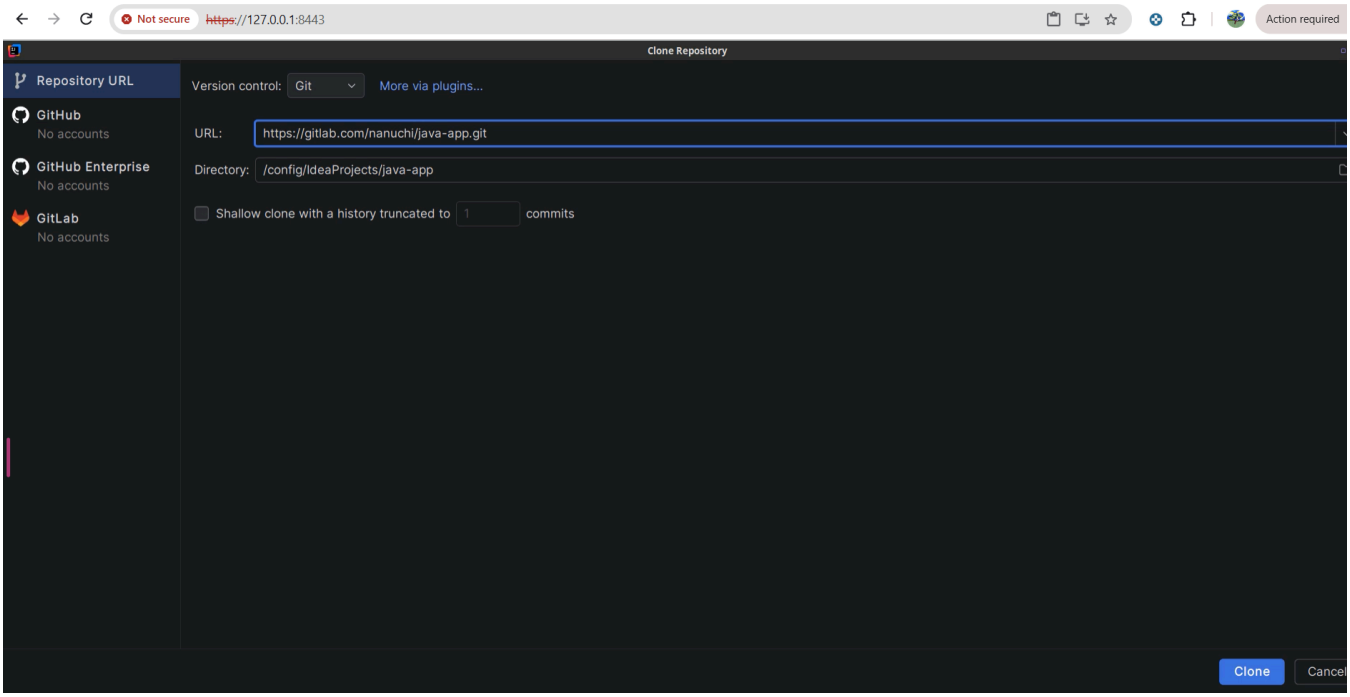
Work performed / key points

- Load the project as a Gradle project and select the Gradle Wrapper.
- Run ./gradlew build and wait for BUILD SUCCESSFUL. If permission is denied, run chmod +x gradlew and retry.

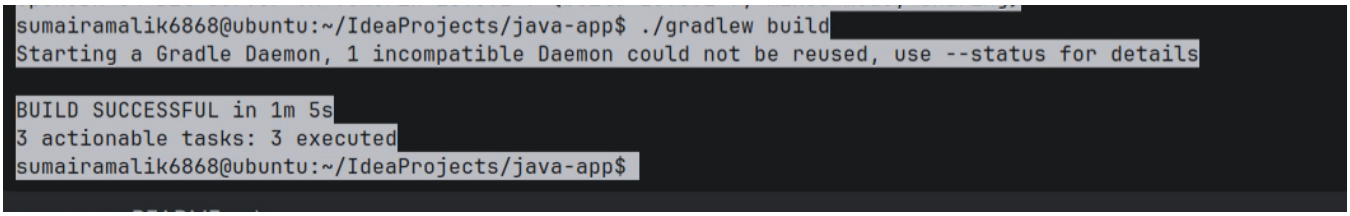
- Add implementation("org.elasticsearch:elasticsearch:7.0.0") inside the existing dependencies block in build.gradle. Synchronize Gradle.
- Run ./gradlew build again and wait for BUILD SUCCESSFUL.
- Locate build/libs/java-app-1.0-SNAPSHOT.jar.
- Run java -jar build/libs/java-app-1.0-SNAPSHOT.jar, stop it with Ctrl+C, and close the project.

Evidence — insert your screenshot below each item

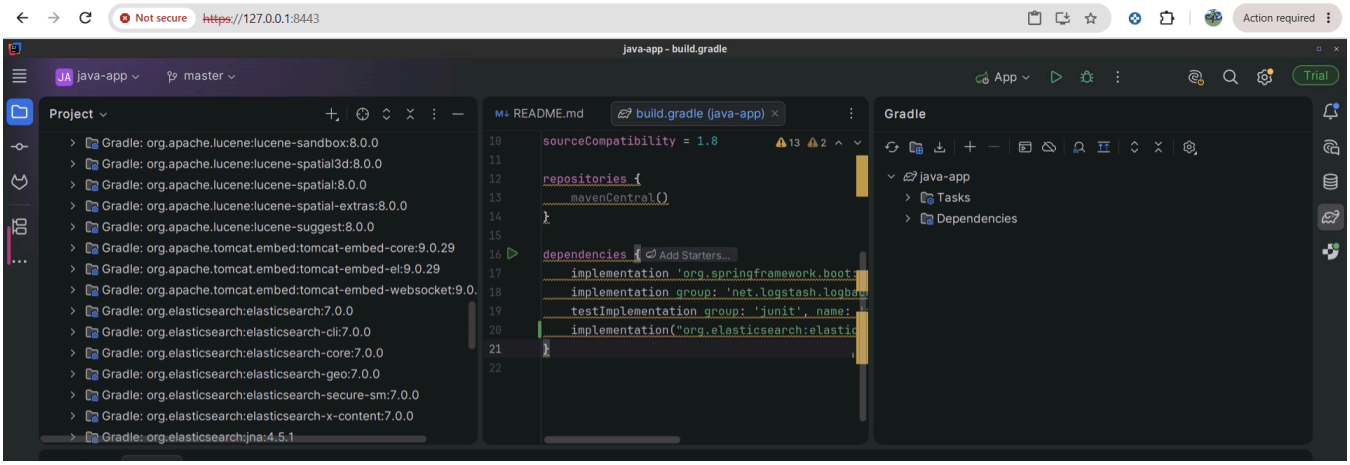
gradle_project_clone.png — Clone dialog showing the Gradle repository URL and destination.



gradle_build.png — Initial Gradle Wrapper build and BUILD SUCCESSFUL.



gradle_elasticsearch_dependency.png — build.gradle dependency and synchronized Elasticsearch library.



gradle_rebuild.png — Successful build after adding Elasticsearch.

```
sumairamalik6868@ubuntu:~/IdeaProjects/java-app$ ./gradlew build
Starting a Gradle Daemon, 1 incompatible Daemon could not be reused, use --status for details

BUILD SUCCESSFUL in 1m 5s
3 actionable tasks: 3 executed
```

gradle_jar.png — build/libs directory and exact JAR filename.

```
sumairamalik6868@ubuntu:~/IdeaProjects/java-app$ ls -lh build/libs/
total 18M
-rw-r--r-- 1 abc abc 18M Nov  3 17:31 java-app-1.0-SNAPSHOT.jar

sumairamalik6868@ubuntu:~/IdeaProjects/java-app$
```

gradle_running.png — JAR command and successful application startup.

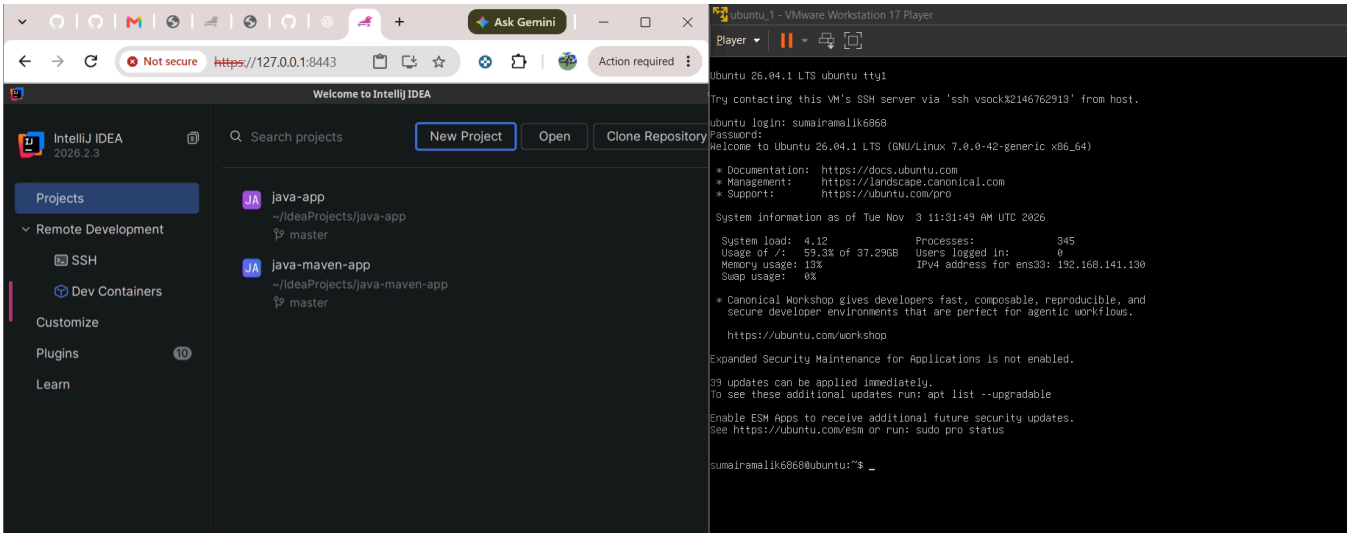
```
sumairamalik6868@ubuntu:~/IdeaProjects/java-app$ java -jar build/libs/java-app-1.0-SNAPSHOT.jar
{"app":"JavaApp","time":"2026-11-03T13:03:02.501+00:00","logger":"com.example.App","level":"INFO","class":"org.springframework.boot.StartupInfoLogger","method":"logStarting","file"
:"StartupInfoLogger.java","line":55,"thread":"main","msg":"Starting App on ubuntu with PID 6149 (/config/IdeaProjects/java-app/build/libs/java-app-1.0-SNAPSHOT.jar started by abc i
n /config/IdeaProjects/java-app)"}
{"app":"JavaApp","time":"2026-11-03T13:03:02.523+00:00","logger":"com.example.App","level":"INFO","class":"org.springframework.boot.SpringApplication","method":"logStartupProfileIn
fo","file":"SpringApplication.java","line":651,"thread":"main","msg":"No active profile set, falling back to default profiles: default"}
{"app":"JavaApp","time":"2026-11-03T13:03:04.664+00:00","logger":"org.apache.coyote.http11.Http11NioProtocol","level":"INFO","class":"org.apache.juli.logging.DirectJDKLog","method"
:"log","file":"DirectJDKLog.java","line":173,"thread":"main","msg":"Initializing ProtocolHandler [\"http-nio-8080\"]"}
{"app":"JavaApp","time":"2026-11-03T13:03:04.665+00:00","logger":"org.apache.catalina.core.StandardService","level":"INFO","class":"org.apache.juli.logging.DirectJDKLog","method":"
log","file":"DirectJDKLog.java","line":173,"thread":"main","msg":"Starting service [Tomcat]"}
{"app":"JavaApp","time":"2026-11-03T13:03:04.665+00:00","logger":"org.apache.catalina.core.StandardEngine","level":"INFO","class":"org.apache.juli.logging.DirectJDKLog","method":"l
og","file":"DirectJDKLog.java","line":173,"thread":"main","msg":"Starting Servlet engine: [Apache Tomcat/9.0.29]"}
{"app":"JavaApp","time":"2026-11-03T13:03:04.753+00:00","logger":"org.apache.catalina.core.ContainerBase.[Tomcat].[localhost].[/]","level":"INFO","class":"org.apache.juli.logging.D
irectJDKLog","method":"log","file":"DirectJDKLog.java","line":173,"thread":"main","msg":"Initializing Spring embedded WebApplicationContext"}
{"app":"JavaApp","time":"2026-11-03T13:03:04.818+00:00","logger":"com.example.App","level":"INFO","class":"com.example.App","method":"init","file":"App.java","line":21,"thread":"ma
in","msg":"Java app started"}
{"app":"JavaApp","time":"2026-11-03T13:03:05.239+00:00","logger":"org.apache.coyote.http11.Http11NioProtocol","level":"INFO","class":"org.apache.juli.logging.DirectJDKLog","method"
:"log","file":"DirectJDKLog.java","line":173,"thread":"main","msg":"Starting ProtocolHandler [\"http-nio-8080\"]"}
{"app":"JavaApp","time":"2026-11-03T13:03:05.285+00:00","logger":"com.example.App","level":"INFO","class":"org.springframework.boot.StartupInfoLogger","method":"logStarted","file":
"StartupInfoLogger.java","line":61,"thread":"main","msg":"Started App in 3.922 seconds (JVM running for 5.743)"}
```

gradle_stopped.png — Ctrl+C / shutdown output and returned terminal prompt.

```
{"app":"JavaApp","time":"2026-11-03T13:03:05.239+00:00","logger":"org.apache.coyote.http11.Http11NioProtocol
:"log","file":"DirectJDKLog.java","line":173,"thread":"main","msg":"Starting ProtocolHandler [\"http-nio-808
{"app":"JavaApp","time":"2026-11-03T13:03:05.285+00:00","logger":"com.example.App","level":"INFO","class":"o
"StartupInfoLogger.java","line":61,"thread":"main","msg":"Started App in 3.922 seconds (JVM running for 5.74

^Csumairamalik6868@ubuntu:~/IdeaProjects/java-app$
```

gradle_project_closed.png — IntelliJ Welcome screen after closing the Gradle project.



Task 4 — Install and Run the Node.js Project

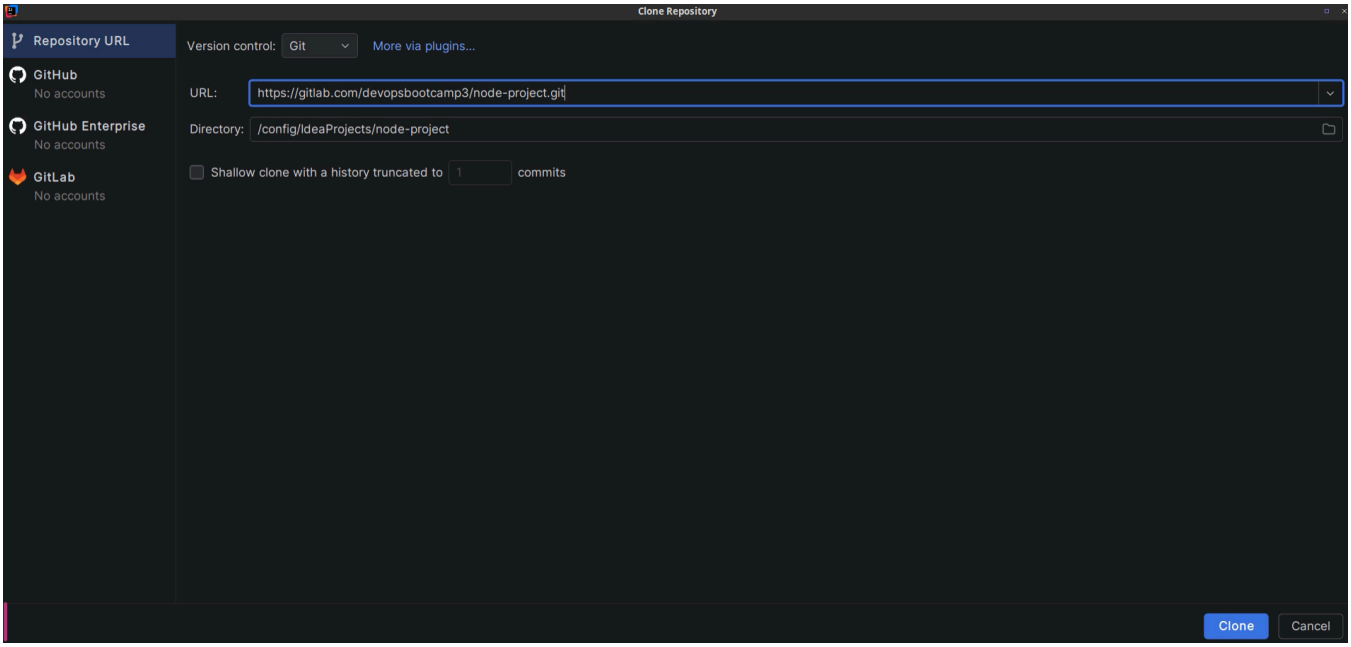
The Node.js sample is cloned from https://gitlab.com/devopsbootcamp3/node-project.git into /config/IdeaProjects/node-project. Its application files are in the app directory.

Work performed / key points

- Install nodejs and npm in IntelliJ's terminal and verify with node --version and npm --version.
- Enter the app directory and run npm install to install dependencies.
- Run npm start. If port 3000 reports EADDRINUSE, change the listener and its message in app/server.js to port 4000, save, and run npm start again.
- While the application is running, open a browser inside the container desktop at http://localhost:4000 and verify the project page.
- Stop the app with Ctrl+C. In the app directory run npm test, then npm pack. The package should be bootcamp-node-project-1.0.0.tgz.

Evidence — insert your screenshot below each item

node_project_clone.png — Clone dialog showing the Node.js repository URL and destination.



nodejs_installed.png — Installed Node.js and npm versions.

```
sumairamalik6868@ubuntu:~/IdeaProjects/node-project$ sudo apt install -y nodejs
Installing:
  nodejs

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
  Download size: 37.8 MB
  Space needed: 236 MB / 13.1 GB available

Get:1 https://deb.nodesource.com/node_22.x nodistro/main amd64 nodejs amd64 22.23.4-1nodesource1 [37.8 MB]
Fetched 37.8 MB in 46s (819 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79, <STDIN> line 1.)
debconf: falling back to frontend: Readline
Selecting previously unselected package nodejs.
(Reading database ... 52313 files and directories currently installed.)
Preparing to unpack .../nodejs_22.23.4-1nodesource1_amd64.deb ...
Unpacking nodejs (22.23.4-1nodesource1) ...
Setting up nodejs (22.23.4-1nodesource1) ...
Processing triggers for man-db (2.13.1-1build1) ...
sumairamalik6868@ubuntu:~/IdeaProjects/node-project$ node -v
npm -v
v22.23.4
10.9.9
sumairamalik6868@ubuntu:~/IdeaProjects/node-project$
```

npm_install.png — Completed npm install in the app directory.

```
npm warn deprecated is-accessor-descriptor@0.1.6: Please upgrade to v0.1.7
npm warn deprecated is-data-descriptor@0.1.4: Please upgrade to v0.1.5
npm warn deprecated is-data-descriptor@0.1.4: Please upgrade to v0.1.5
npm warn deprecated is-accessor-descriptor@0.1.6: Please upgrade to v0.1.7
npm warn deprecated is-data-descriptor@0.1.4: Please upgrade to v0.1.5
npm warn deprecated is-accessor-descriptor@0.1.6: Please upgrade to v0.1.7
npm warn deprecated is-accessor-descriptor@0.1.6: Please upgrade to v0.1.7
npm warn deprecated is-data-descriptor@0.1.4: Please upgrade to v0.1.5
npm warn deprecated is-data-descriptor@0.1.4: Please upgrade to v0.1.5
npm warn deprecated is-accessor-descriptor@0.1.6: Please upgrade to v0.1.7

added 554 packages, and audited 555 packages in 2m

24 packages are looking for funding
  run `npm fund` for details

54 vulnerabilities (10 low, 20 moderate, 20 high, 4 critical)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 10.9.9 -> 12.2.1
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.2.1
npm notice To update run: npm install -g npm@12.2.1
npm notice

sumairamalik6868@ubuntu:~/IdeaProjects/node-project/app$
```

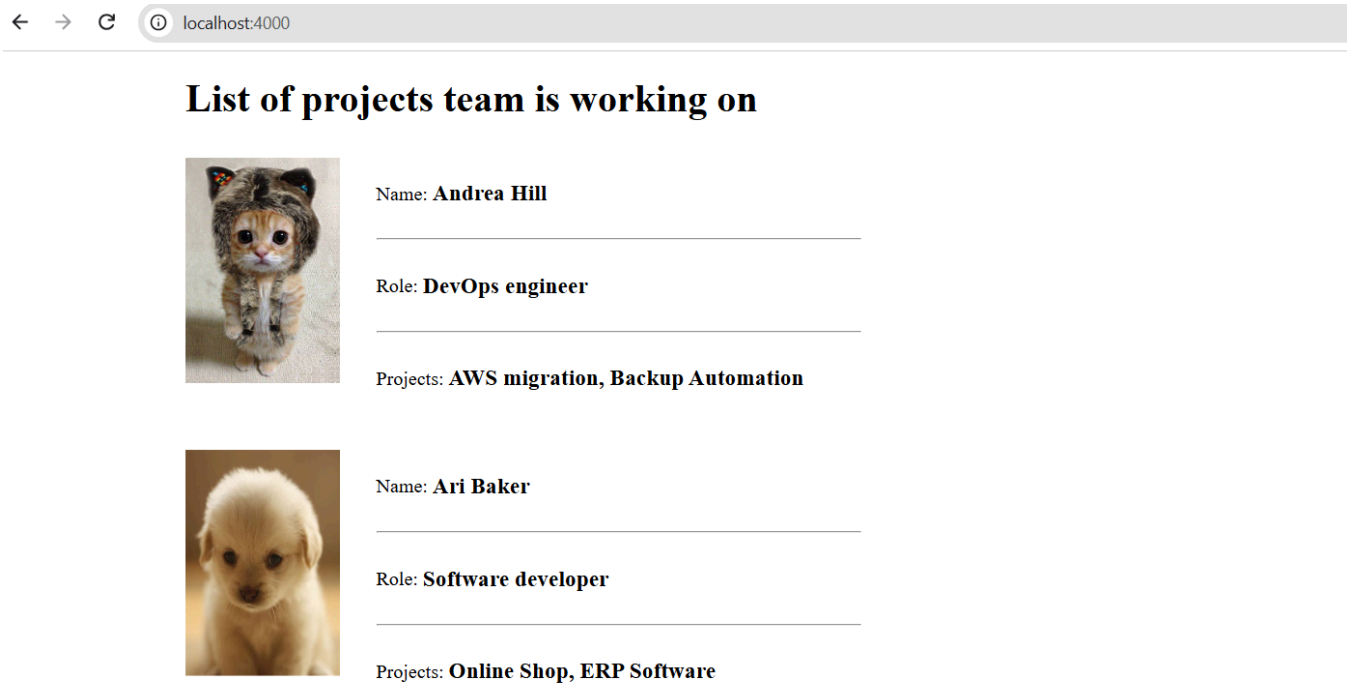
node_running.png — npm start output showing app listening on port 4000.

```
sumairamalik6868@ubuntu:~/IdeaProjects/node-project/app$ npm start

> bootcamp-node-project@1.0.0 start
> node server.js

app listening on port 4000!
```

node_browser.png — Application page at http://localhost:4000 in the container browser.



npm_test.png — npm test command and passing test summary.

```
sumairamalik6868@ubuntu:~/IdeaProjects/node-project/app$ npm test

> bootcamp-node-project@1.0.0 test
> jest

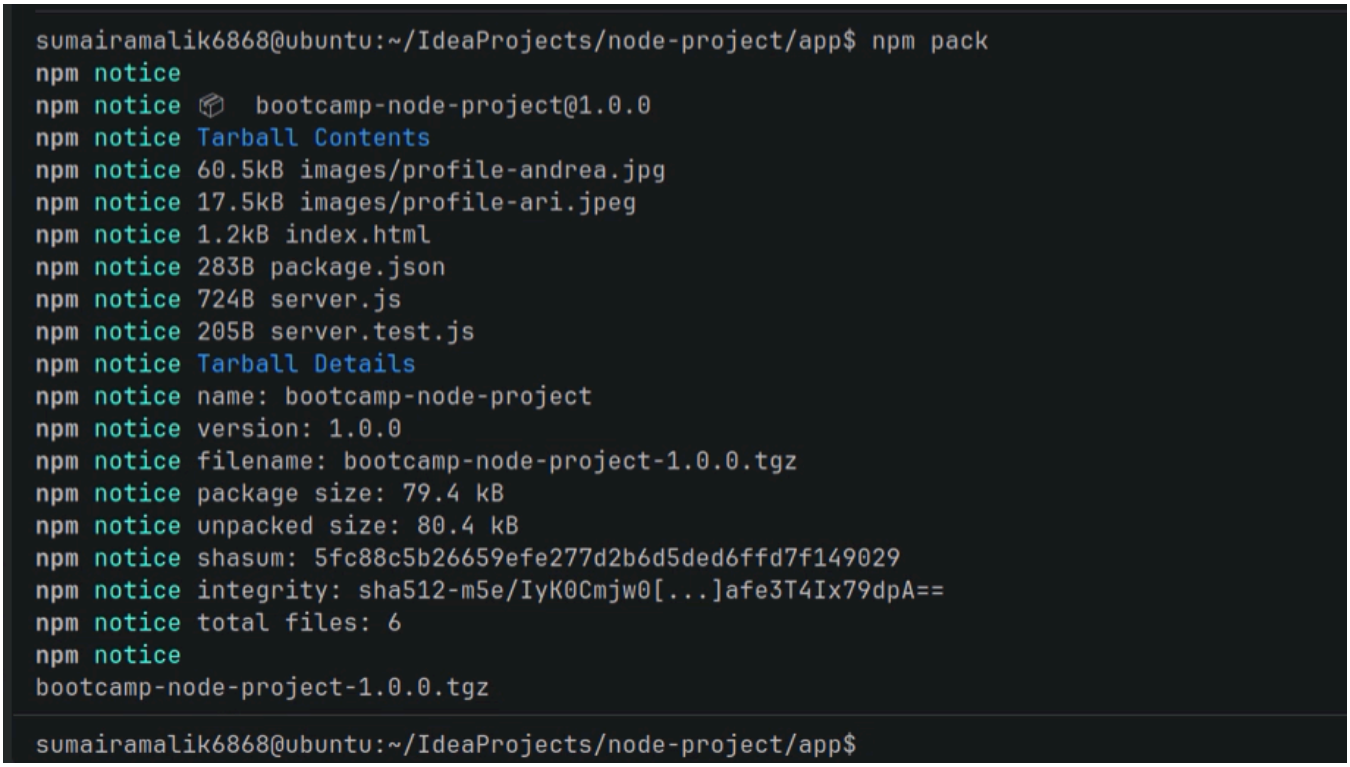
Browserslist: caniuse-lite is outdated. Please run:
npx browserslist@latest --update-db

Why you should do it regularly:
https://github.com/browserslist/browserslist#browsers-data-updating
 PASS  ./server.test.js
  ✓ main index.html file exists (11 ms)

Test Suites: 1 passed, 1 total
Tests:       1 passed, 1 total
Snapshots:   0 total
Time:        5.741 s
Ran all test suites.

sumairamalik6868@ubuntu:~/IdeaProjects/node-project/app$
```

npm_pack.png — npm pack output and generated .tgz filename.



Task 5 — Build the React and Node.js API Artifact

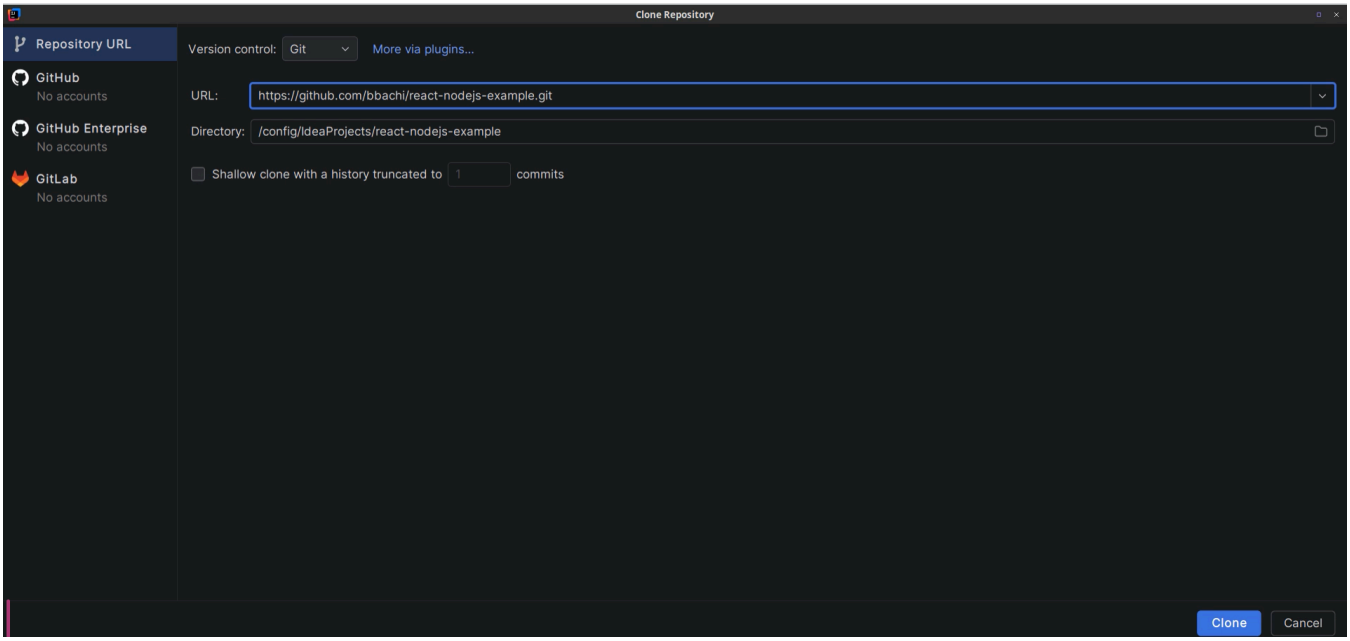
The React/Node.js example is cloned from https://github.com/bbachi/react-nodejs-example.git into /config/IdeaProjects/react-nodejs-example. This task builds the Node.js API located in the api directory.

Work performed / key points

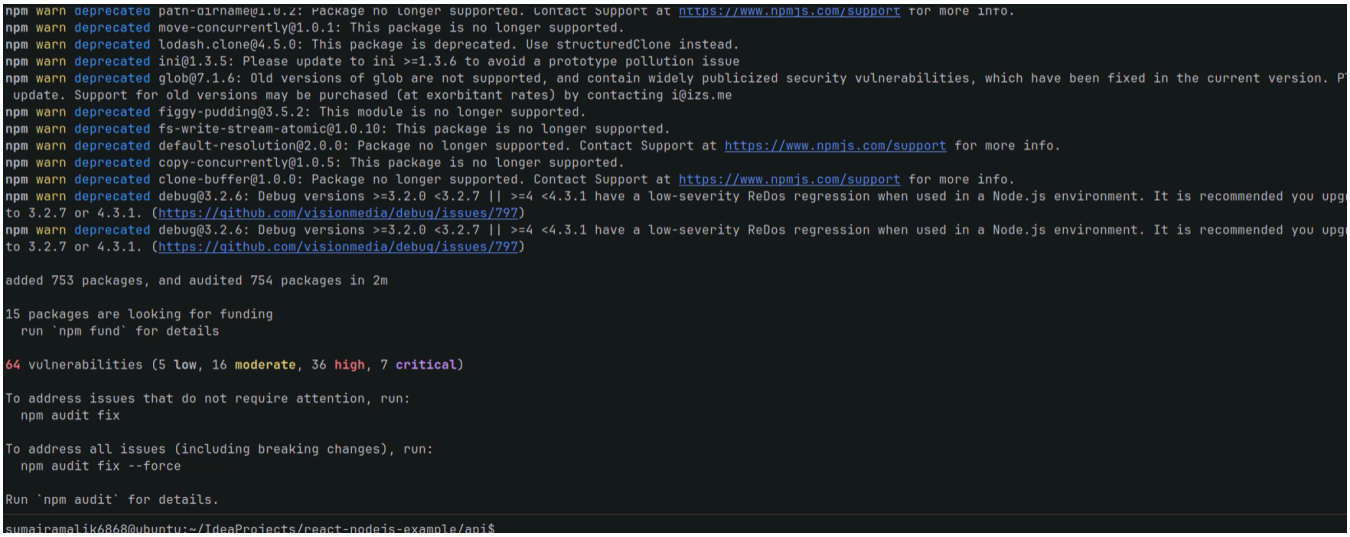
- From the project root, enter api and run npm install. Verify api/node_modules in the Project pane.
- Run npm run build. The webpack configuration generates api/server.bundle.js.
- If the older webpack build reports ERR_OSSL_EVP_UNSUPPORTED, retry with NODE_OPTIONS=--openssl-legacy-provider npm run build.
- View the generated bundle with cat server.bundle.js | more. Press Space to move through the output and q to quit.

Evidence — insert your screenshot below each item

react_node_project_clone.png — Clone dialog showing repository URL and destination.



react_api_dependencies.png — Completed npm install and api/node_modules directory.



react_api_build.png — Successful webpack build and server.bundle.js in api.

```
sumairamalik6868@ubuntu:~/IdeaProjects/react-nodejs-example/my-app$ export NODE_OPTIONS=--openssl-legacy-provider
npm run build

> my-app@0.1.0 build
> react-scripts build

Creating an optimized production build...
Browserslist: caniuse-lite is outdated. Please run:
npx browserslist@latest --update-db
Compiled with warnings.

./src/components/DisplayBoard.js
  Line 5:11:  'headerStyle' is assigned a value but never used  no-unused-vars

Search for the keywords to learn more about each warning.
To ignore, add // eslint-disable-next-line to the line before.

File sizes after gzip:

  42.41 KB  build/static/js/2.67710b93.chunk.js
  22.51 KB  build/static/css/2.af3c1da9.chunk.css
  1.94 KB   build/static/js/main.3ca0bfa3.chunk.js
  775 B     build/static/js/runtime-main.7c0ae09a.js
  401 B     build/static/css/main.14ff7981.chunk.css

The project was built assuming it is hosted at /.
You can control this with the homepage field in your package.json.

The build folder is ready to be deployed.
You may serve it with a static server:

  npm install -g serve
```

react_api_bundle.png — Bundle viewing command and bundled JavaScript contents.

```
sumairamalik6868@ubuntu:~/IdeaProjects/react-nodejs-example/my-app$ tar -czvf react-nodejs-example.tar.gz build/
build/
build/index.html
build/static/
build/static/css/
build/static/css/main.14ff7981.chunk.css
build/static/css/main.14ff7981.chunk.css.map
build/static/css/2.af3c1da9.chunk.css.map
build/static/css/2.af3c1da9.chunk.css
build/static/js/
build/static/js/runtime-main.7c0ae09a.js
build/static/js/runtime-main.7c0ae09a.js.map
build/static/js/2.67710b93.chunk.js.LICENSE.txt
build/static/js/2.67710b93.chunk.js
build/static/js/main.3ca0bfa3.chunk.js.map
build/static/js/main.3ca0bfa3.chunk.js
build/static/js/2.67710b93.chunk.js.map
build/asset-manifest.json
build/favicon.ico
build/logo192.png
build/precache-manifest.e2425b7827f4dd6bab1b61c7d8569038.js
build/manifest.json
build/logo512.png
build/robots.txt
build/service-worker.js

sumairamalik6868@ubuntu:~/IdeaProjects/react-nodejs-example/my-app$ ls -lh react-nodejs-example.tar.gz
-rw-r--r-- 1 abc abc 304K Nov  4 10:27 react-nodejs-example.tar.gz

sumairamalik6868@ubuntu:~/IdeaProjects/react-nodejs-example/my-app$
```

Task 6 — Build a Docker Image for the Node.js Project

Reopen the existing node-project clone from Task 4. Run the Docker commands from /config/IdeaProjects/node-project, where the Dockerfile is located.

Work performed / key points

- Verify Docker with docker --version and confirm the Dockerfile with ls.
- List existing images with sudo docker images.
- Build the image with sudo docker build -t lab5:1.0 . and wait for a successful build.

- Run sudo docker images again and verify repository lab5 and tag 1.0. Together they identify lab5:1.0.

Evidence — insert your screenshot below each item

docker_installed.png — Docker version output.

```
sumairamalik6868@ubuntu:~/IdeaProjects/react-nodejs-example$ docker --version
Docker version 29.9.2, build a82e528

sumairamalik6868@ubuntu:~/IdeaProjects/react-nodejs-example$
```

dockerfile_exists.png — ls output showing Dockerfile.

```
sumairamalik6868@ubuntu:~/IdeaProjects/react-nodejs-example$ ls -l Dockerfile
-rw-r--r-- 1 abc abc 347 Nov  4 09:57 Dockerfile

sumairamalik6868@ubuntu:~/IdeaProjects/react-nodejs-example$
```

docker_images_before.png — Docker image list before building.

```
sumairamalik6868@ubuntu:~/IdeaProjects/node-project/app$ sudo docker images

IMAGE                                  ID             DISK USAGE   CONTENT SIZE   EXTRA
intellij-idea-docker-intellij:latest   d69188a01dd8       10.9GB         2.85GB   U

sumairamalik6868@ubuntu:~/IdeaProjects/node-project/app$ cd ..
```

docker_build.png — Docker build command and successful build output.

```
sumairamalik6868@ubuntu:~/IdeaProjects/node-project$ sudo docker build -t lab5:1.0 .
[+] Building 168.1s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 156B
 => [internal] load metadata for docker.io/library/node:13-alpine
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load build context
 => => transferring context: 37.92MB
 => [1/5] FROM docker.io/library/node:13-alpine@sha256:527c70f74817f6f6b5853588c28de33459178ab72421f1fb7b63a281ab670258
 => => resolve docker.io/library/node:13-alpine@sha256:527c70f74817f6f6b5853588c28de33459178ab72421f1fb7b63a281ab670258
 => => sha256:5d74fb112a7d313344f9f679ccb975062377464235fac36e1b952aa2b125d5e2 2.24MB / 2.24MB
 => => sha256:4b9536424fa1675675e7edd59ae9a92573f0c010c5458733a493e2c3e1daf1ae 283B / 283B
 => => sha256:780514bed1adda642a7f7008ea5b1477bb78a23e3f76eabbdde21031b2185a2f 35.30MB / 35.30MB
 => => sha256:cbdbe7a5bc2a134ca8ec91be58565ec07d037386d1f1d8385412d224deafca08 2.81MB / 2.81MB
 => => extracting sha256:cbdbe7a5bc2a134ca8ec91be58565ec07d037386d1f1d8385412d224deafca08
 => => extracting sha256:780514bed1adda642a7f7008ea5b1477bb78a23e3f76eabbdde21031b2185a2f
 => => extracting sha256:5d74fb112a7d313344f9f679ccb975062377464235fac36e1b952aa2b125d5e2
 => => extracting sha256:4b9536424fa1675675e7edd59ae9a92573f0c010c5458733a493e2c3e1daf1ae
 => [2/5] COPY app/ /usr/app/
 => [3/5] WORKDIR /usr/app
 => [4/5] RUN npm install
 => [5/5] RUN npm run test
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:425b338eb1db1e24946de29c3a4f759d70abcc05616cb77610cf1352f11b98c4
 => => exporting config sha256:e464f763e90708f85a66cbbd94c1b466c1a64b34f6321d19d6757e37b0998afe
 => => exporting attestation manifest sha256:c8688f57e0b361a19f918d969f8bf1d94eecc293dcbc000c29ad1bb7e8816b55
 => => exporting manifest list sha256:850be65f44041db44fc901f9af44ef726c41e9340fc67677715c813dd3befac6
 => => naming to docker.io/library/lab5:1.0
 => => unpacking to docker.io/library/lab5:1.0
```

docker_images_after.png — Docker image list showing lab5 repository, 1.0 tag, and image ID.

```
sumairamalik6868@ubuntu:~/IdeaProjects/node-project$ sudo docker images

IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
intellij-idea-docker-intellij:latest d69188a01dd8       10.9GB         2.85GB    U
lab5:1.0                             850be65f4404        417MB         85.1MB

sumairamalik6868@ubuntu:~/IdeaProjects/node-project$
```

Exam Evaluation Questions (Practice Only)

1. Maven built java-maven-app-1.1.0-SNAPSHOT.jar. Which values in pom.xml determine this name, and what does the package phase do?

The JAR filename is determined by the artifactId, version, and packaging type in pom.xml. In this sample, the artifactId is java-maven-app, the version is 1.1.0-SNAPSHOT, and the packaging is JAR. The Maven package phase runs the earlier lifecycle phases needed to compile and process the project, then packages the compiled application into a JAR in the target directory.

2. Why can ./gradlew build work without a separate system-wide Gradle installation? Which JDK settings would you check if this sample reports an unsupported Java version?

The Gradle Wrapper uses the Gradle version configured for the project and can download that version when needed, so a separate system-wide Gradle installation is not required. If an unsupported Java version error occurs, check the Project SDK, the Gradle JVM setting in IntelliJ, JAVA_HOME, and the Java version supported by the project's Gradle Wrapper.

3. Why does EADDRINUSE occur on port 3000 in this environment, and how does changing the application listener to port 4000 let the container's browser open it?

EADDRINUSE means the requested port is already in use by another process. In this environment, port 3000 is used by the desktop service, so the sample application cannot bind to it. Changing the application's listener to port 4000 avoids that port conflict. The application can then be opened from the container desktop at http://localhost:4000.

Overall Lab Summary

In this lab, the supplied IntelliJ Docker environment was launched and four sample projects were opened. Elasticsearch 7.0.0 was added to the Maven and Gradle projects and both were rebuilt. The Java applications were packaged as JAR files, executed, stopped, and their project windows closed. Node.js dependencies were installed, the application was started on port 4000, its tests were run, and an npm tarball was created. The React/Node.js API dependencies were installed and webpack generated server.bundle.js. Finally, the Node.js project was built into the Docker image lab5:1.0 and the image was verified.

Official References

- IntelliJ Docker setup repository: https://github.com/WaqasSaleem97/Intellij-Idea-docker
- Maven sample project: https://gitlab.com/nanuchi/java-maven-app
- Gradle sample project: https://gitlab.com/nanuchi/java-app
- Node.js sample project: https://gitlab.com/devopsbootcamp3/node-project
- React/Node API package.json: https://github.com/bbachi/react-nodejs-example/blob/master/api/package.json
- React/Node webpack configuration: https://github.com/bbachi/react-nodejs-example/blob/master/api/webpack.config.js
- JetBrains — Clone a repository: https://www.jetbrains.com/help/idea/set-up-a-git-repository.html
- JetBrains — Maven dependencies: https://www.jetbrains.com/help/idea/work-with-maven-dependencies.html
- JetBrains — Gradle projects: https://www.jetbrains.com/help/idea/work-with-gradle-projects.html
- JetBrains — IntelliJ terminal: https://www.jetbrains.com/help/idea/terminal-emulator.html
- JetBrains — SDK configuration: https://www.jetbrains.com/help/idea/sdk.html
- Maven — Build lifecycle: https://maven.apache.org/guides/introduction/introduction-to-the-lifecycle.html
- Gradle — Java compatibility: https://docs.gradle.org/current/userguide/compatibility.html
- Gradle — Declaring dependencies: https://docs.gradle.org/current/userguide/declaring_dependencies_basics.html
- Gradle — Wrapper: https://docs.gradle.org/current/userguide/gradle_wrapper.html
- npm — Install: https://docs.npmjs.com/cli/v11/commands/npm-install/
- npm — Test: https://docs.npmjs.com/cli/v11/commands/npm-test/
- npm — Pack: https://docs.npmjs.com/cli/v11/commands/npm-pack/
- Node.js — OpenSSL compatibility: https://nodejs.org/en/blog/release/v17.0.0#openssl-30
- Docker — Build image: https://docs.docker.com/reference/cli/docker/buildx/build/
- Docker — List images: https://docs.docker.com/reference/cli/docker/image/ls/
- npm — Start: https://docs.npmjs.com/cli/v11/commands/npm-start/
- LinuxServer Webtop: https://docs.linuxserver.io/images/docker-webtop/